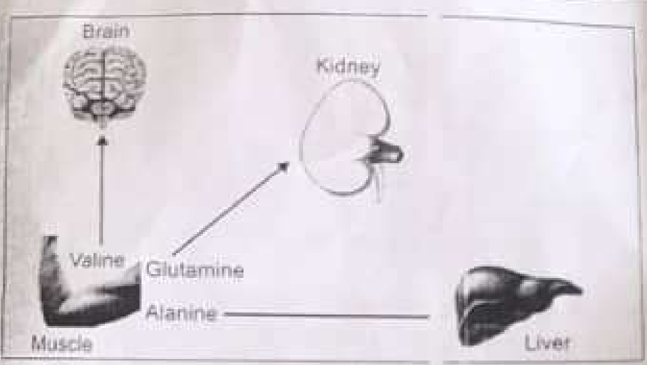


Fig. 17.2: Interorgan transport of amino acids during fasting conditions

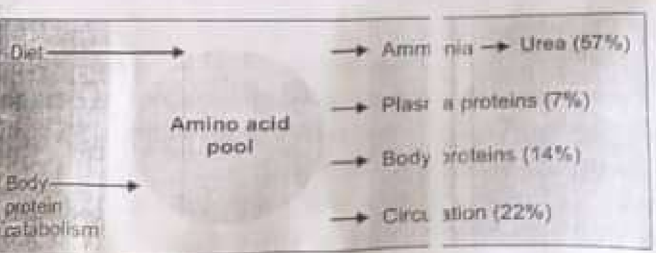


Fig. 17.4: Amino acid pool

because it is seen in all cells abundantly. It is a small protein with 76 residues (molecular weight, 8.5 kDa). Ubiquitin is attached to proteins.

Proteasomes

Ubiquitin tagged proteins are immediately broken down inside the **proteasomes** of the cells. Ciechanover, Hershko and Rose were awarded Nobel Prize in 2004 for their discovery of ubiquitin-mediated protein degradation.

Interorgan Transport of Amino Acids

Breakdown of muscle protein is the source of amino acids for tissues while liver is the site of disposal (Figs. 17.2 and 17.3).

In Fasting State

The muscle releases mainly alanine and glutamine of which **alanine is taken up by liver** and glutamine by kidneys (Fig. 17.2). Liver removes the amino group and converts it to urea and the carbon skeleton is used for **gluconeogenesis**. Students should also see glucose-alanine cycle, under gluconeogenesis (*see* Fig. 10.27). The brain predominantly takes up branched chain amino acids.

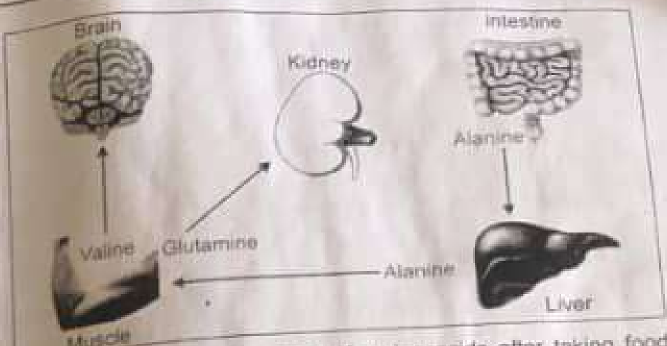


Fig. 17.3: Interorgan transport of amino acids after taking food (postprandial condition)

In the Fed State

Amino acids absorbed from the diet are taken up by different tissues (Fig. 17.3). Both muscle and brain take up branched chain amino acids, and release glutamine and alanine. The glutamine is delivered to kidneys to aid in regulation of acid-base balance, while alanine is taken up by liver.

General Metabolism of Amino Acids

1. The **anabolic reactions** where proteins are synthesized.
2. **Synthesis** of specialized products such as heme, creatine, purines and pyrimidines.
3. The **catabolic reactions** where dietary proteins and body proteins are broken down to amino acids.
4. **Transamination**: Amino group is removed to produce the carbon skeleton (keto acid). The amino group is excreted as **urea**.
5. The carbon skeleton is used for synthesis of **non-essential** amino acids.
6. It is also used for **gluconeogenesis** or for complete oxidation.
7. Amino acids are used for minor metabolic functions like conjugation, methylation, amidation, etc. Figure 17.1 gives a summary of amino acid metabolism. The amino acid pool in the body is shown in Figure 17.4.

FORMATION OF AMMONIA

The sources and fate of ammonia are shown in Figure 17.5. The first step in the catabolism of amino acids is to remove the amino group as **ammonia**. This is the major source of ammonia. However, small quantities of ammonia may also be formed from catabolism of purine and pyrimidine bases.

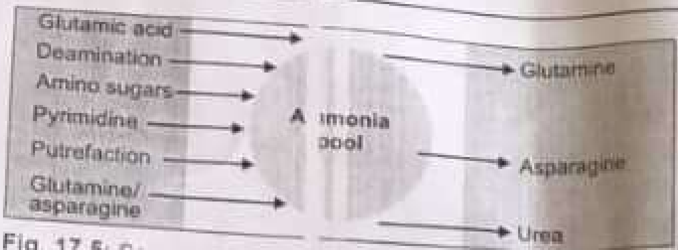


Fig. 17.5: Sources and fate of ammonia

Ammonia is **highly toxic** especially to the nervous system. Detoxification of ammonia is by conversion to urea and excretion through urine.

Transamination

Transamination is the exchange of the alpha amino group between one alpha amino acid and another alpha keto acid, forming a new alpha amino acid.

Amino acid 1 + keto acid 2 → amino acid 2 + keto acid 1

As an example, amino groups are interchanged between alanine and glutamic acid (Fig. 17.6). In almost all cases, the amino group is accepted by **alpha ketoglutaric acid** so that glutamic acid is formed. The enzymes catalyzing the reaction as a group are known as **aminotransferases**. These enzymes have **pyridoxal phosphate** as prosthetic group (Fig. 17.6). The reaction is readily reversible.

Biological Significance of Transamination

First Step of Catabolism

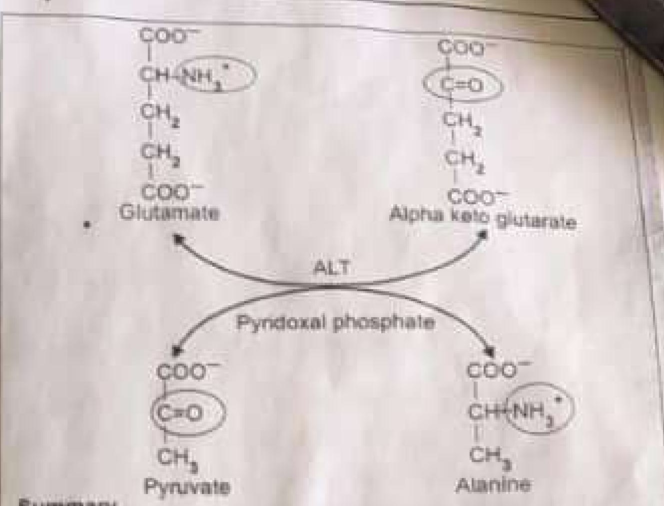
In this first step, **ammonia** is removed, and the carbon skeleton of the amino acid enters into catabolic pathway.

Synthesis of Non-essential Amino Acids

By means of transamination, all nonessential amino acids can be synthesized by the body from keto acids available from other sources. For example, pyruvate can be transaminated to synthesize **alanine**. Similarly oxaloacetate produces aspartic acid. Alpha ketoglutarate is transaminated to form glutamic acid. Those amino acids, which cannot be synthesized in this manner, are therefore essential; they should be made available in the food (*See* Box 3.1 for essential amino acids).

Interconversion of Amino Acids

If amino acid no.1 is high and no. 2 is low, the amino group from no.1 may be transferred to a keto acid to give amino acid no. 2 to equalize the quantity of both. This is called **equalization** of quantities of non-essential amino acids.



Summary
amino acid 1 + keto acid 2 → amino acid 2 + keto acid 1

Fig. 17.6: Transamination reaction. In this example, enzyme is Alanine aminotransferase (ALT) and pyridoxal phosphate is the coenzyme. The reaction is readily reversible

Exceptions

Lysine, threonine and proline are not transaminated. They follow direct degradative pathways.

Clinical Significance of Transamination

Aspartate amino transferase (**AST**) and Alanine amino transferase (**ALT**) are induced by glucocorticoids, which favor gluconeogenesis. AST is increased in **myocardial infarction** and ALT in **liver** diseases. Their clinical importance is given in Chapter 6.

Transdeamination

1. The amino group of most of the amino acids is released by a coupled reaction, transdeamination, that is **transamination followed by oxidative deamination**.
2. Transamination takes place in the cytoplasm of all the cells of the body; the amino group is transported to liver as **glutamic acid,** which is finally oxidatively deaminated in the mitochondria of hepatocytes.
3. Thus, the two components of the reaction are physically far away, but physiologically they are coupled. Hence, the term transdeamination (Fig. 17.7).

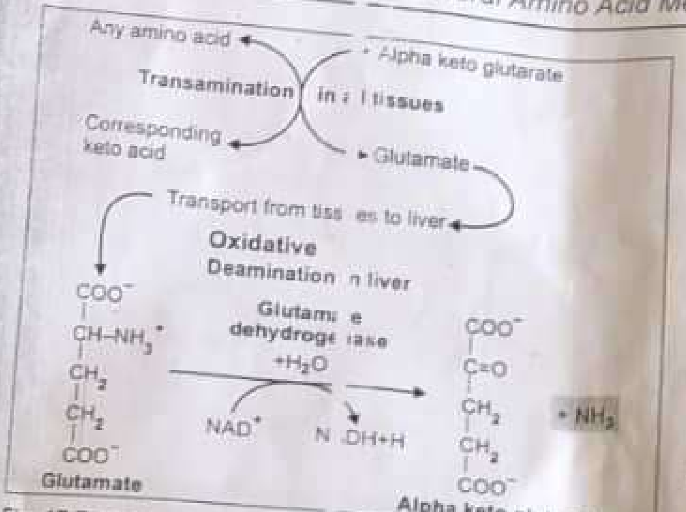


Fig. 17.7: Transamination + deamination = transdeamination

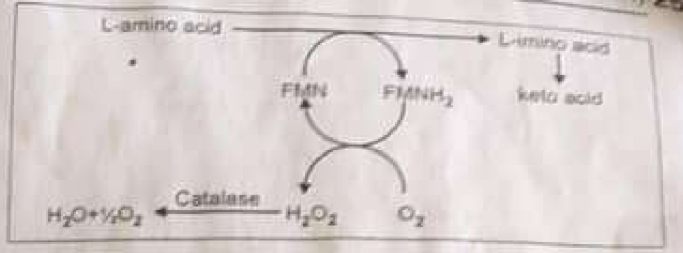


Fig. 17.8: L-amino acid oxidase

Oxidative Deamination of Glutamate

Only **liver** mitochondria contain **glutamate dehydrogenase (GDH)** which deaminates glutamate to alpha-ketoglutarate plus ammonia. So, all amino acids are first transaminated to glutamate, which is then finally deaminated **(transdeamination)** (Fig.17.7). Amino acids are deaminated at the rate of about 50–70 gram per day.

During the transamination reaction the amino group of all other amino acids is funneled into glutamate. Hence, the glutamate dehydrogenase reaction is the final reaction, which removes the amino group of all amino acids (Fig. 17.7). It needs NAD^+ as coenzyme. It is an allosteric enzyme; it is activated by ADP and inhibited by GTP.

The hydrolysis of glutamine also yields NH_3 but this occurs mainly in the kidney where the NH_4^+ excretion is required for acid-base regulation.

Minor Pathways of Deamination

1. L-amino acid oxidase can act on all amino acids except hydroxy amino acids and dicarboxylic amino acids. It uses FMN as coenzyme. The peroxide formed in this reaction is decomposed by catalase in the peroxisomes (Fig. 17.8).
2. D-amino acid oxidase can oxidize glycine and any D amino acid that may be formed by bacterial metabolism. It uses FAD as coenzyme.
3. Small quantities of ammonia may be formed in the body through minor reactions like oxidation of monoamines by MAO (mono amine oxidase) (*see* Chapter 19, under Tyrosine metabolism).

Nonoxidative Deaminations

Dehydratases act on hydroxy amino acids to remove ammonia from the following amino acids:

a. **Serine** will give rise to pyruvate (*see* Chapter 18).
b. **Threonine** is converted to alpha keto butyric acid.

Desulfhydrase: Cysteine undergoes deamination and simultaneous transsulfuration to form pyruvate (*see* Chapter 18).

Histidine also undergoes non-oxidative deamination to form urocanic acid; catalyzed by histidase (*see* Chapter 19). Ammonia may also be produced by degradation of purines and pyrimidines due to bacterial putrefaction in the gastrointestinal tract.

DISPOSAL/DETOXIFICATION OF AMMONIA

First Line of Defense (Trapping of Ammonia)

Being highly toxic, ammonia should be eliminated or detoxified, as and when it is formed. Even very minute quantity of ammonia may produce toxicity in central nervous system. But, ammonia is always produced by almost all cells, including neurons. The intracellular ammonia is immediately trapped by glutamic acid to form **glutamine**, especially in brain cells (Fig. 17.9). The glutamine is then transported to liver, where the reaction is reversed by the enzyme **glutaminase** (Fig. 17.9). The ammonia thus generated is immediately detoxified into urea. Aspartic acid may also undergo similar reaction to form **asparagine** (*see* Chapter 18).

Transportation of Ammonia

Inside the cells of almost all tissues, the transamination of amino acids produce glutamic acid. However, glutamate dehydrogenase is available only in the liver. Therefore, the final deamination and production of ammonia

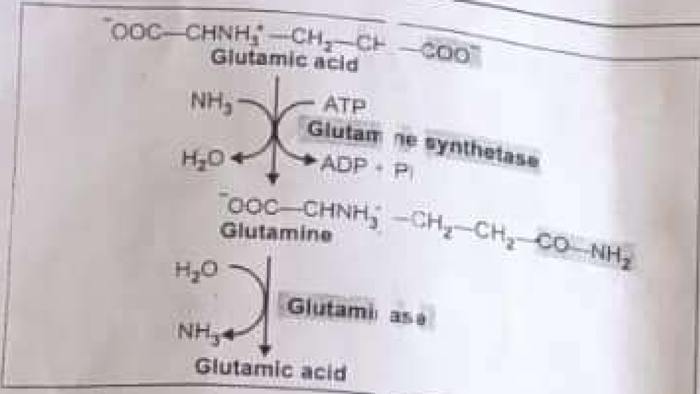


Fig. 17.9: Ammonia trapping as glutamine

BOX 17.3: Mammals excrete ammonia as urea; but birds excrete ammonia as uric acid

Millions of "gooney" birds nest on some islands of Pacific Ocean, off the coast of Peru. Over the centuries, their droppings formed big hills. These "guano" deposits, containing mainly uric acid, is now being exploited commercially as fertilizer containing nitrogen.

is taking place in the liver (Fig. 17.7). Thus, **glutamic acid** acts as the link between amino groups of amino acids and ammonia. The concentration of glutamic acid in blood is 10 times more than other amino acids. **Glutamine** is the transport form of ammonia from brain and intestine to liver; while alanine is the transport form from muscle to liver.

Final Disposal

The ammonia from all over the body thus reaches liver. It is then **detoxified to urea by liver** cells, and excreted through kidneys. **Urea is the end product of protein metabolism.**

Since mammals including human beings excrete amino nitrogen mainly as urea, they are referred to as ureotelic. Fishes excrete ammonia as such (ammoneotelic) while birds and reptiles excrete nitrogen as uric acid (uricotelic) (Box 17.3).

Although, ammonia is toxic and has to be immediately detoxified, in kidney cells, ammonia is generated from glutamine by the action of glutaminase. The purpose is to excrete hydrogen ions as ammonium ions, so as to maintain acid-base balance (*see* Fig. 27.5).

UREA CYCLE

In 1773, Rouelle isolated urea from urine. Frederic Wohler in 1828 obtained urea by boiling an aqueous

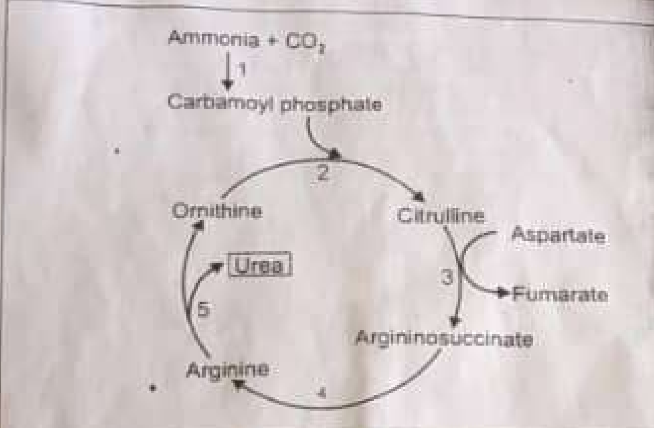


Fig. 17.10: Urea cycle, summary

Table 17.2: Comparison, CPS-I and II enzymes			
		CPS-I	CPS-II
1.	Site	Mitochondria	Cytosol
2.	Pathway of	Urea	Pyrimidine
3.	Positive effector	NAG	Nil
4.	Source for N	Ammonia	Glutamine
5.	Inhibitor	Nil	CTP

solution of ammonium cyanate. The urea cycle is the first metabolic pathway to be elucidated in 1932, by Hans Krebs (1900–1981) and Kurt Henseleit (1907–1972). Hence, the cycle is known as Krebs-Henseleit urea cycle. As ornithine is the first member of the reaction, it is also called as Ornithine cycle.

The two nitrogen atoms of urea are derived from two different sources, one from ammonia and the other directly from the alpha amino group of aspartic acid.

Step 1: Formation of Carbamoyl Phosphate

One molecule of ammonia condenses with CO_2 in the presence of **two molecules of ATP** to form carbamoyl phosphate. The reaction is catalyzed by the mitochondrial enzyme **carbamoyl phosphate synthetase-I (CPS-I)**, (Figs. 17.10 and 17.11, Step 1). An entirely different cytoplasmic enzyme, carbamoyl phosphate synthetase-II, (CPS-II) is involved in pyrimidine nucleotide synthesis (*see* Chapter 38). The differences of CPS-I and II are shown in Table 17.2. CPS-I reaction is the **rate-limiting step** in urea formation. It is irreversible and allosterically regulated.

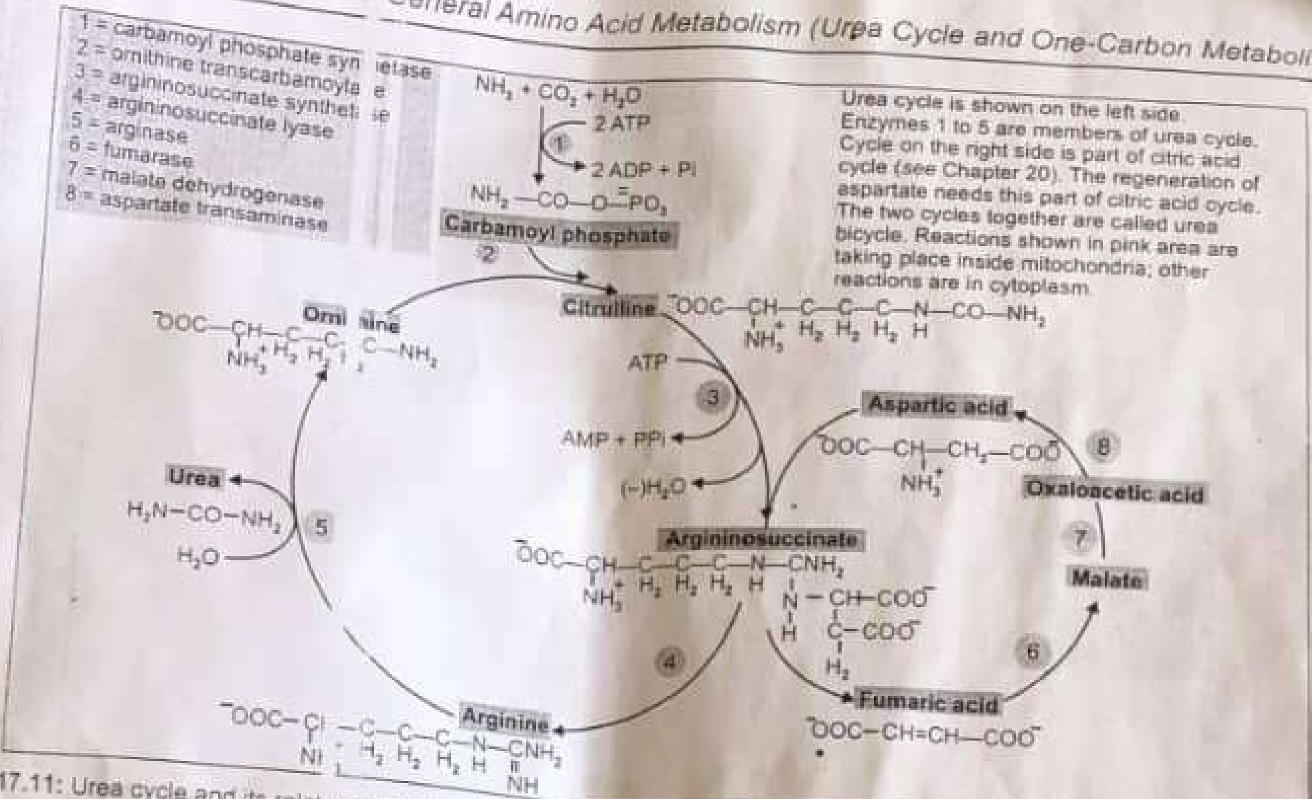


Fig. 17.11: Urea cycle and its relation with citric acid cycle

Step 2: Formation of Citrulline

The second reaction is also **mitochondrial.** The carbamoyl group is transferred to the NH_2 group of ornithine by **ornithine transcarbamoylase** (OTC) (Figs. 17.10 and 17.11, Step 2). The citrulline leaves the mitochondria and further reactions are taking place in cytoplasm. Citrulline is neither present in tissue proteins nor in blood; but it is **present in milk**.

Step 3: Formation of Argininosuccinate

One molecule of aspartic acid adds to citrulline forming a carbon to nitrogen bond which provides the 2nd nitrogen atom of urea. **Argininosuccinate synthetase** catalyzes the reaction (Figs. 17.10 and 17.11, Step 3). This needs hydrolysis of ATP to AMP level, so **two high energy phosphate bonds** are utilized.

Step 4: Formation of Arginine

Argininosuccinate is cleaved by **argininosuccinate lyase** (argininosuccinase) to arginine and fumarate (Figs. 17.10 and 17.11, Step 4). The enzyme is inhibited by fumarate. But this is avoided by the cytoplasmic localization of the enzyme. The fumarate formed may be funnelled into TCA cycle to be converted to malate and then to oxaloacetate to be transaminated to aspartate (Fig. 17.11). Thus the urea cycle is linked to TCA cycle through fumarate. The 3rd and 4th steps taken together may be summarized as:

Citrulline + aspartate → Arginine + fumarate

A similar reaction of donation of amino group by aspartate takes place in purine nucleotide synthesis also (*see* Chapter 38).

Step 5: Formation of Urea

The final reaction of the cycle is the hydrolysis of arginine to urea and ornithine by arginase (Figs. 17.10 and 17.11, Step 5). The ornithine returns to the mitochondria to react with another molecule of carbamoyl phosphate so that the cycle is repeated. Thus ornithine may be considered as a catalyst which enters the reaction and is regenerated.

Energetics of Urea Cycle

The overall reaction may be summarized as:

NH_3 + CO_2 + Aspartate → Urea + Fumarate

In the urea cycle 2 ATPs are used in the first reaction. Another ATP is converted to AMP and PPi, which is